

Houru Jiang

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Guangzhou, China

Research Profile

I work on **evaluation-centred trustworthy AI** and **data-centric LLM systems**. My research develops mathematical and model-agnostic methods – including compression, entropy, causal intervention, and uncertainty estimation – for efficient data selection, benchmark and model diagnosis, and AI system auditing. I am also interested in efficient LLM training and architectures, including mathematically grounded pre-training and post-training pipelines and alternatives such as linear attention.

Core Interests:

Efficient and interpretable LLM data selection; mathematical methods for pre-training and post-training; information-theoretic evaluation; causal model auditing; efficient LLM training and inference; linear attention and other efficient LLM architectures.

Selected Manuscripts and Submissions

- **CDIS: Coupling Compression Ratio and Token Entropy for Training-Free Instruction Selection**

Houru Jiang, Zixi Wang, Mingliang Hou, Xueyi Li, Teng Guo, Jiaqi Zheng, Renqiang Luo
AAAI 2027 Conference Submission. First Author.

- **HB++: Topology-Conditioned Propagation Bases for Shallow Graph Learning**

Houru Jiang, Zixi Wang, Mingliang Hou, Tengting Cheng, Zitao Liu, Teng Guo, Jiaqi Zheng
WSDM 2027 Conference Submission. First Author.

- **An Information-Theoretic Evaluation Framework for Benchmark and Model Diagnosis in Knowledge Tracing**

Houru Jiang*, Zixi Wang*, Tengting Cheng, Xueyi Li, Mingliang Hou, Jiaqi Zheng, Renqiang Luo, Teng Guo, Zitao Liu
NeurIPS 2026 submission. Co-First Author.

- **ResidKT: Disentangling Sequential Dynamics via Neuro-Symbolic Prior-Residual Learning for Knowledge Tracing**

Zixi Wang*, Houru Jiang*, Tengting Cheng, Xueyi Li, Mingliang Hou, Zitao Liu
AAAI 2027 Conference Submission. Co-First Author.

- **VLM-CD: Making Concept Annotations Optional with Frozen Vision-Language Embeddings**

Zixi Wang, Houru Jiang, Mingliang Hou, Youheng Bai, Renqiang Luo, Teng Guo, Jiaqi Zheng, Qianru Liang
AAAI 2027 Conference Submission. Second Author.

- **All-as-Vision: A Content-Native Exercise Interface for Transferable Knowledge Tracing**

Zixi Wang, Qing Qing, Houru Jiang, Mingliang Hou, Renqiang Luo, Xikun Zhang, Ziqi Xu
AAAI 2027 Conference Submission. Third Author.

* Equal contribution / co-first authors.

Research Experience

Research Assistant

Nov 2024 – Sep 2025

Professor Hao Chen, Macau University of Science and Technology (Remote)

- Conducted research on knowledge graphs, graph representation learning, recommendation systems, and knowledge-enhanced retrieval.
- Analyzed how structured knowledge and retrieval signals can support recommendation and information-access systems, contributing to data organization, method implementation, and experimental validation.

Research Assistant

May 2025 – Present

Associate Professor Mingliang Hou

Jinan University

- Designed CDIS, a CPU-only instruction-data selector that couples compression ratio and token entropy for low-cost, interpretable LLM data selection.
- Developed an information-theoretic framework for localized benchmark and model diagnosis in knowledge tracing.
- Built causal and probabilistic auditing methods to test whether graph-based models genuinely utilize their input graph structures.
- Contributed to a robotic-manipulation project through system evaluation, experimental analysis, and reproducible comparison studies.

Education

Jinan University

Guangzhou, China

Bachelor of Information Management and Information Systems, Department of Mathematics

Expected Graduation: Jun 2027

GPA: 3.34/5.00; Average: 83.4/100

Relevant Coursework

Mathematics: Mathematical Analysis I–III, Advanced Algebra I–II, Probability and Statistics, Discrete Mathematics, Operations Research

Computer Science: Data Structures, Object-Oriented Programming, Database Systems, Computer Networks

Artificial Intelligence: Computational Intelligence, Advanced Artificial Intelligence

Technical Skills

Programming and Scientific Computing: Python, MATLAB, SQL, Java, C, Pandas, NumPy, Matplotlib, Seaborn

Machine Learning and LLM Systems: PyTorch, Scikit-Learn, Transformers, Hugging Face, Qwen, DeepSeek, OpenAI API, Ollama, Stable-Baselines3

Tools: Linux, Git, Jupyter, LaTeX

Honors and Awards

- Jinan University Third-Class Outstanding Student Scholarship
- Mathematical Contest in Modeling (MCM/ICM), Successful Participant (2024)
- Mathematical Contest in Modeling (MCM/ICM), Honorable Mention (2025)

Language Proficiency

IELTS: Overall 7.0

Listening 7.5 | Reading 7.5 | Speaking 6.0 | Writing 6.0